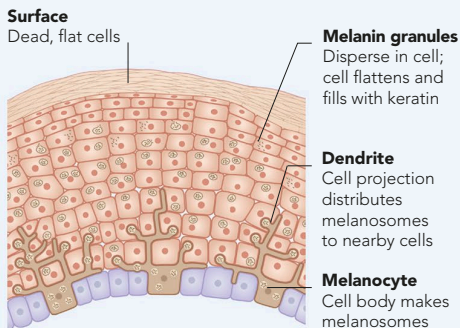


ULTRAVIOLET DEFENCES

The Sun's rays include a spectrum of colour wavelengths, as well as infrared or IR rays and ultraviolet, UV, rays. Both UV-A and UV-B wavelengths are invisible, but exposure to the latter, in particular, is linked to skin cancers. Skin's defence is its dark pigment, melanin. This forms a screen in the upper epidermis and shields the multiplying cells in the base of the epidermis.

MELANIN PRODUCTION

Melanocytes are melanin-producing cells in the base of the epidermis. They make parcels of melanin granules, known as melanosomes, which pass into surrounding cells.



SKIN PIGMENTATION

Skin colour depends on the type and quantity of two melanin pigments – reddish pheomelanin and brown-black eumelanin – in the epidermis, and on the way the pigment granules are distributed. Each melanocyte has finger-like dendrites that touch surrounding cells (basal keratinocytes). The melanocyte produces

pigment granules within organelles called melanosomes. These move along the dendrites and into nearby cells. Darker skin has larger melanocytes with more melanosomes. Lighter skin has smaller melanocytes and fewer melanosomes. Exposure to UV rays stimulates the melanocytes so the skin darkens.

COLOUR VARIATION

Darker skin tends to have larger melanin-making cells with denser melanosomes, in comparison to lighter skin. The former release their pigment granules, while the latter's granules stay clumped.

